$$
\begin{aligned}
\mathrm{Corr}(Z_1, Z_2) &= \frac{\mathrm{Cov}(Z_1, Z_2)}{\sqrt{\mathrm{Var}(Z_1)\mathrm{Var}(Z_2)}} \\
&= \mathrm{Cov}(Z_1, Z_2) \\
&= \mathrm{E}\big((Z_1 - \mu_{Z_1})(Z_2 - \mu_{Z_2})\big) \\
&= E(Z_1 Z_2) \\
&= 0
\end{aligned}
$$

$$
\mathrm{E}(Y_2) = \rho_{12}\mathrm{E}(Z_1) + \sqrt{1-\rho_{12}^2}\,\mathrm{E}(Z_2) = 0
$$

$$
\mathrm{Var}(Y_2) = \rho_{12}^2\mathrm{Var}(Z_1) + (1-\rho_{12}^2)\mathrm{Var}(Z_2) = \rho_{12}^2 + 1 - \rho_{12}^2 = 1
$$

$$
\begin{aligned}
\mathrm{Corr}(Y_1, Y_2) &= \frac{\mathrm{Cov}(Y_1, Y_2)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_2)}} \\
&= \mathrm{Cov}(Y_1, Y_2) \\
&= \mathrm{Cov}\left(Z_1, \rho_{12}Z_1 + \sqrt{1-\rho_{12}^2}\,Z_2\right) \\
&= \rho_{12}\mathrm{Var}(Z_1) + \sqrt{1-\rho_{12}^2}\,\mathrm{Cov}(Z_1, Z_2) \\
&= \rho_{12}
\end{aligned}
$$

$$
\begin{aligned}
\rho_{13} &= \mathrm{Corr}(Y_1, Y_3) \\
&= \frac{\mathrm{Cov}(Y_1, Y_3)}{\sqrt{\mathrm{Var}(Y_1)\mathrm{Var}(Y_3)}} \\
&= \mathrm{Cov}(Y_1, Y_3) \\
&= E(Y_1 Y_3) \\
&= E\big(Z_1(aZ_1 + bZ_2 + cZ_3)\big) \\
&= aE(Z_1^2) + bE(Z_1 Z_2) + cE(Z_1 Z_3) \\
&= a
\end{aligned}
$$

$$
\begin{aligned}
\rho_{23} &= \mathrm{Corr}(Y_2, Y_3) \\
&= \mathrm{Cov}\left(\rho_{12}Z_1 + \sqrt{1-\rho_{12}^2}\,Z_2, aZ_1 + bZ_2 + cZ_3\right) \\
&= \rho_{12}a + \sqrt{1-\rho_{12}^2}\,b
\end{aligned}
$$

$$b = \frac{\rho_{23} - \rho_{12}a}{\sqrt{1-\rho_{12}^2}}$$

$$\begin{aligned}\mathrm{Var}(Y_3) &= \mathrm{Var}(aZ_1 + bZ_2 + cZ_3) \\ &= a^2\mathrm{Var}(Z_1) + b^2\mathrm{Var}(Z_2) + c^2\mathrm{Var}(Z_3) \\ &= a^2 + b^2 + c^2 = 1\end{aligned}$$

$$c = \sqrt{1-a^2-b^2} = \sqrt{1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2}}$$

For $i = 1, 2, 3$

$$\boxed{X_i = \mu_i + \sigma_i Y_i}$$

$$\mathrm{E}(X_i) = \mu_i + \sigma_i \mathrm{E}(Y_i) = \mu_i$$

$$\mathrm{Var}(X_i) = \sigma_i^2 \mathrm{Var}(Y_i) = \sigma_i^2$$

For $i \neq j$

$$\begin{aligned}\mathrm{Corr}(X_i, X_j) &= \mathrm{Corr}(\mu_i + \sigma_i Y_i, \mu_j + \sigma_j Y_j) \\ &= \mathrm{Corr}(Y_i, Y_j) \\ &= \rho_{ij}\end{aligned}$$

This solution was submitted by **Jon**
Published and refined by **Jon**

Last updated: June 13, 2026